আপনি যে ORM গুলো দেখলেন (select_related, prefetch_related, Prefetch, F, Subquery, Exists ইত্যাদি) — এখন সহজভাবে **কোনটা কী করে** তা এক লাইনে বুঝিয়ে দিচ্ছি।

Django ORM – কোনটা কী করে

1 select_related

কাজ:

ForeignKey বা OneToOne relation এর data **SQL JOIN করে এক query তে আনে।**

Example Model

```
class Author(models.Model):
    name = models.CharField(max_length=100)

class Book(models.Model):
    title = models.CharField(max_length=100)
    author = models.ForeignKey(Author, on_delete=models.CASCADE)
```

```
books = Book.objects.select_related('author')
```

✔ Query কমায়

✔ FK relation fast করে

2 prefetch_related

কাজ:

ManyToMany বা Reverse ForeignKey relation এর data **multiple query করে Python এ merge করে।**

Example

```
books = Book.objects.prefetch_related('authors')
```

✔ ManyToMany relation optimize করে

3 Prefetch

কাজ:

prefetch_related এর **advanced version**

Filter / custom queryset ব্যবহার করা যায়।

Example

```
from django.db.models import Prefetch

books = Book.objects.prefetch_related(
    Prefetch(
        'authors',
        queryset=Author.objects.filter(name__startswith='A')
    )
)
```

✔ Filtered prefetch

---

4 F() Expression

কাজ:

Database field দিয়ে **calculation করা**।

Example

```
from django.db.models import F

Product.objects.update(stock=F('stock') - 1)
```

✔ Database level calculation
✔ Fast

---

5 Subquery

কাজ:

একটা query এর ভিতরে **আরেকটা query চালানো**।

Example

```
from django.db.models import Subquery

latest_review = Review.objects.filter(product=OuterRef('pk'))
```

```
Product.objects.annotate(
    latest_rating=Subquery(latest_review.values('rating')[:1])
)
```

✔ Nested query

---

6 Exists

কাজ:

Check করা relation **আছে কিনা।**

Example

```
from django.db.models import Exists

reviews = Review.objects.filter(product=OuterRef('pk'))

Product.objects.annotate(
    has_review=Exists(reviews)
)
```

✔ True / False result

---

7 annotate()

কাজ:

Extra field calculate করে add করা।

Example

```
from django.db.models import Count

Product.objects.annotate(
    review_count=Count('review')
)
```

✔ Aggregation

---

8 values()

কাজ:

Object না এনে **dictionary data আনা।**

Example

Product.objects.values('id', 'name')

Output

{'id':1,'name':'Phone'}

9 values_list()

কাজ:

Database থেকে **tuple / list আনা।**

Example

Product.objects.values_list('name', flat=True)

Output

['Phone', 'Laptop']

bulk_create

কাজ:

একসাথে অনেক object insert করা।

Example

Product.objects.bulk_create(products)

✔ 1 query
✔ Fast insert

সবচেয়ে গুরুত্বপূর্ণ 5টা (Real Project)

ORM	কাজ
select_related	FK relation optimize

ORM	কাজ
prefetch_related	ManyToMany optimize
F()	database calculation
annotate	aggregation
bulk_create	fast insert

যেহেতু আপনি **Django E-commerce project করছেন**, সবচেয়ে বেশি ব্যবহার করবেন:

select_related
prefetch_related
annotate
F()

Django ORM Optimization – Visual Summary

```
┌──────────────────┐
│  select_   │
│  related   │
└──────────┘
    │
    │ FK / OneToOne
    │ SQL JOIN → 1 Query
    ▼
┌──────────────────────────────────┐
│ Book.objects.select_related│
│ ('author')              │
└──────────────────────────────────┘

┌──────────────────┐
│ prefetch_    │
│ related      │
└──────────┘
    │
    │ ManyToMany / Reverse FK
    │ Multiple query + Python merge
    ▼
┌──────────────────────────────────┐
│ Book.objects.prefetch_      │
```

```
│ related('authors')          │
└──────────────────────────────────────────┘

        ┌──────────────────┐
        │  Prefetch    │
        └──────────┬───────┘
                   │
                   │ Advanced prefetch
                   │ filter / custom queryset / alias
                   ▼
┌──────────────────────────────────────────┐
│ Book.objects.prefetch_      │
│ Prefetch('authors',         │
│ queryset=Author.objects... │
│ to_attr='a_authors')        │
└──────────────────────────────────────────┘
```

```
┌──────────────────┐
│    F()      │
└──────────┬───────┘
           │
           │ Database calculation
           ▼
Product.objects.update(stock=F('stock') - 1)

┌──────────────────┐
│  Subquery   │
└──────────┬───────┘
           │
           │ Nested query
           ▼
Product.objects.annotate(latest_rating=Subquery(...))

┌──────────────────┐
│   Exists    │
└──────────┬───────┘
           │
           │ Check relation exists
           ▼
Product.objects.annotate(has_review=Exists(...))
```

```
┌──────────────┐
│ annotate     │
└──────┬───────┘
       │
       │ Aggregation
       ▼
Product.objects.annotate(review_count=Count('review'))

┌──────────────┐
│ values() /   │
│ values_list  │
└──────┬───────┘
       │
       │ Dictionary / List data
       ▼
Product.objects.values('id', 'name')
Product.objects.values_list('name', flat=True)

┌──────────────┐
│ bulk_create  │
└──────┬───────┘
       │
       │ Fast insert multiple objects
       ▼
Product.objects.bulk_create([...])
```

Quick Rules

ORM Function	Use Case
select_related	FK / OneToOne, SQL JOIN
prefetch_related	ManyToMany / Reverse FK, Python merge
Prefetch	Filter / custom queryset / alias
F()	Database calculation / update
Subquery	Nested query / annotate
Exists	Relation check (True/False)

ORM Function	Use Case
annotate	Aggregation / extra field
values() / values_list	Lightweight dictionary / list data
bulk_create	Insert many objects in 1 query

ORM Function	Relation Type / Use Case	Query Type	কেমন Query হয়	Notes
select_related	ForeignKey / OneToOne	Single SQL query	JOIN ব্যবহার করে একসাথে data আনে	1 query, N+1 problem ফ্রী
prefetch_related	ManyToMany / Reverse FK	Multiple queries	2 বা বেশি query, পরে Python memory তে merge করে	FK না, Python merge
Prefetch	ManyToMany / Reverse FK + Filter / alias	Multiple queries	Filtered / customized query, Python merge	Advanced prefetch
F()	Field calculation / update	Single SQL update	UPDATE table SET field = field + 1	Python এ load হয় না
Subquery	Nested query	Single query (with subquery)	SELECT ..., (SELECT ...) AS sub	Annotate বা filter এর জন্য
Exists	Check relation exists	Single query (with EXISTS)	SELECT ..., EXISTS(SELECT 1 ...) AS has_relation	True / False
annotate()	Aggregation	Single query (with GROUP	SELECT ..., COUNT(...) AS	Add extra field

ORM Function	Use Case			
		BY)	total	
values()	Lightweight data	Single query	SELECT field1, field2 FROM table	Object না নিয়ে dict
values_list()	Lightweight list / tuple	Single query	SELECT field1, field2 FROM table	Flat=True হলে list
bulk_create	Insert many objects	Single SQL insert	INSERT INTO table (fields) VALUES (...)	Python loop ছাড়াই fast
bulk_update	Update many objects	Single SQL update	UPDATE table SET field=val WHERE ...	Python loop ছাড়া